MIDTERM PROJECT

AI System Development and Experimental Safety Evaluation

Course: AI Security and Risk Analysis

Course Code: AISRA1301

Programme: Master's Degree — AI+X

Institution: Qazaq AI Research University (QAIRU)

Instructor: Professor Dr. Tamara Zhukabayeva

1. Project Overview

The Midterm Project is a continuation of Assignment 2, in which students reviewed and presented a scientific research paper related to AI safety, security, robustness or risk analysis.

For the Midterm, students must move from theoretical analysis to practical implementation.

Each student is required to develop a functional AI prototype, implement or adapt a relevant approach from the selected research paper, and conduct an initial experimental safety or security evaluation.

The project must demonstrate practical implementation, scientific reasoning, experimental evidence and an understanding of AI-related risks.

Important: Students are not required to develop a new AI model from scratch or invent a completely new algorithm. The focus is on practical implementation, experimental evaluation and a clearly justified research extension.

The Midterm prototype will serve as the foundation for subsequent assignments and the Final Research Project.

2. Learning Objectives

Upon completion of the Midterm Project, students should be able to:

1. Translate a scientific research problem into a practical AI application or experimental system.
2. Develop and demonstrate a functional AI prototype.
3. Identify relevant AI security, safety or robustness risks.
4. Select and apply an appropriate AI safety evaluation tool.
5. Conduct a controlled and reproducible experiment.
6. Compare experimental results using relevant metrics.
7. Propose and evaluate a research extension or improvement.
8. Explain technical decisions, experimental limitations and the use of AI-assisted development tools.

3. Project Requirements

Part 1. Scientific Paper and Research Problem

Students must continue working with the scientific paper selected in Assignment 2.

They must identify:

- Research title, authors and publication venue.
- Main scientific problem.
- Proposed methodology or algorithm.
- AI system or application considered in the paper.

- Relevant security, safety or robustness risks.
- Limitations of the original research.
- An opportunity for practical implementation or further investigation.

Expected outcome: A clearly defined research problem and an implementation plan based on the selected paper.

Part 2. Functional AI Prototype

Students must develop a working AI application, model or experimental testbed related to the selected research problem.

Possible project types include:

- LLM-based chatbot.
- Retrieval-Augmented Generation (RAG) application.
- AI agent with tool integration.
- Computer vision application.
- Machine learning classification or prediction system.
- AI safety evaluation framework.
- Adversarial machine learning testbed.
- Explainable AI application.

The prototype must:

9. Have a clearly defined purpose.
10. Include a functional AI component.
11. Accept input and produce observable results.
12. Allow the selected safety or security experiment to be conducted.
13. Be demonstrable through a web interface, application, script or notebook.
14. Include sufficient documentation to explain its operation.

Students may use existing open-source models, pretrained models, APIs and software frameworks.

A complex user interface or public deployment is not required.

Part 3. AI-Assisted Development and Vibe Coding

Students are permitted to use AI-assisted development tools to accelerate implementation.

Examples include:

- ChatGPT.
- GitHub Copilot.
- Cursor.
- Claude.
- Google Colab with AI assistance.
- Other suitable coding assistants.

Vibe Coding is permitted, provided that students understand, test and verify the generated code.

Students must be able to explain:

- The architecture of the developed system.
- The role of the AI model.
- The main algorithms and software components.
- How the code was tested.
- Which parts were generated or modified using AI assistance.
- How the correctness and security of the implementation were verified.

AI-generated code without student understanding, testing or verification is not sufficient.

Part 4. AI Safety and Security Experiment

Students must conduct at least one controlled experiment evaluating a relevant safety, security or robustness risk.

Possible research directions include:

A. LLM Security

- Prompt injection.
- Jailbreak resistance.
- Unsafe output generation.
- Sensitive information disclosure.
- RAG document manipulation.

B. Adversarial Machine Learning

- Adversarial examples.
- Robustness under input perturbations.
- Model evasion.
- Classification reliability.

C. AI Agents

- Unsafe tool invocation.
- Excessive permissions.
- Agent instruction manipulation.
- Unintended autonomous actions.

D. AI Reliability and Safety

- Hallucinations.
- Model errors.
- Robustness under changing inputs.
- Safety constraint violations.

All experiments must be performed on systems that students own or are explicitly authorized to test.

Part 5. Research and Evaluation Tools

Students must use at least one appropriate AI safety, security or evaluation tool.

Recommended tools:

Tool	Research Application	Official Link
NVIDIA garak	LLM vulnerability scanning	https://github.com/NVIDIA/garak
Microsoft PyRIT	AI red teaming	https://github.com/Azure/PyRIT
Promptfoo	LLM testing and red teaming	https://www.promptfoo.dev/
IBM ART	Adversarial machine learning	https://github.com/Trusted-AI/adversarial-robustness-toolbox
Giskard	AI and LLM testing	https://github.com/Giskard-AI/giskard
Inspect AI	AI evaluation	https://inspect.aisi.org.uk/
SHAP	Model explainability	https://shap.readthedocs.io/
Ragas	RAG evaluation	https://docs.ragas.io/

The selected tool must be technically appropriate for the chosen AI system.

Students may propose an alternative research tool with appropriate justification.

Part 6. Experimental Design and Results

Students must prepare a reproducible experimental procedure.

The experiment must include:

15. **Research Question:** What specific safety or security problem is being investigated?
16. **Experimental Setup:** Which model, application, dataset and tools are used?

17. **Test Scenarios:** What inputs, attack cases or failure conditions are evaluated?
18. **Baseline:** What is the initial behavior or performance of the system?
19. **Evaluation Metrics:** How are the results measured?
20. **Experimental Results:** What was observed?
21. **Interpretation:** What do the findings demonstrate?
22. **Limitations:** What remains uncertain or untested?

Examples of relevant metrics:

- Attack Success Rate (ASR).
- Detection rate.
- Accuracy.
- Precision, Recall and F1-score.
- Robust accuracy.
- Error rate.
- Hallucination rate.
- Response latency.
- Task success rate.

Students should select metrics appropriate to their research problem.

Experimental claims must be supported by actual results.

Part 7. Research Extension or Improvement

Each student must demonstrate at least one meaningful extension beyond simply running an existing example.

Acceptable extensions include:

- Comparing two models.
- Evaluating an additional attack scenario.
- Implementing a basic safety mitigation.
- Adapting the method to another dataset.
- Comparing baseline and protected configurations.
- Adding a new evaluation metric.
- Investigating a limitation identified in the original paper.
- Evaluating the system under different experimental conditions.

The extension must be clearly identified and justified.

A completely novel algorithm is not required at the Midterm stage.

Part 8. Experimental Comparison

Students must present a comparison table based on their actual results.

Example — LLM Prompt Injection Evaluation

Evaluation Metric	Baseline	Modified Configuration
Test cases	20	20
Successful attacks	8	3
Attack Success Rate	40%	15%
Average response latency	1.2 s	1.5 s

The values above are illustrative only. Students must report their own experimental measurements.

Students must discuss whether the modification improved safety and whether it introduced performance or usability trade-offs.

4. Midterm Deliverables

Students must submit the following materials.

A. Working AI Prototype

A functional implementation of the selected AI system or experimental testbed.

B. Source Code

Provide a GitHub repository or ZIP archive containing:

- Source code or notebooks.
- Dependency information.
- Configuration instructions.
- README.
- Instructions for reproducing the experiment.

C. Technical Research Report

Length: 2–3 pages.

Format: PDF.

Language: English.

Required structure:

23. Project Title.
24. Research Problem and Selected Paper.
25. Proposed AI System.
26. System Architecture.
27. Safety or Security Experiment.
28. Experimental Results.
29. Research Extension.
30. Limitations and Next Steps.
31. References.

D. Presentation

Prepare **6–8 slides** covering:

32. Research Problem and Scientific Paper.
33. Proposed AI System.
34. System Architecture.
35. Research Tool and Experimental Setup.
36. Safety or Security Test.
37. Experimental Results.
38. Research Extension and Contribution.
39. Conclusion and Next Steps.

E. Experimental Evidence

Include relevant evidence such as:

- Test logs.
- Screenshots.
- Evaluation tables.
- Plots.
- Configuration files.
- Notebook outputs.

F. AI Use Declaration

Declare:

- AI tools used.
- Purpose of their use.
- Substantive generated contributions.
- Verification and testing performed by the student.

5. Midterm Assessment Rubric

Assessment Criterion	Points
Scientific paper analysis and research problem	15
Functional AI prototype and implementation quality	20
AI safety or security testing	25
Research extension or improvement	15
Experimental results and critical analysis	15
Demonstration and oral defense	10
Total	**100**

6. Midterm Demonstration and Defense

Each student must demonstrate:

40. The developed AI system.
41. The principal system components.
42. The safety or security testing procedure.
43. The experimental results.
44. The research extension or improvement.
45. The main limitations and planned future work.

Students must be able to explain the code, selected tools, research methodology and interpretation of the results.

7. Academic Integrity and Research Ethics

Students must:

- Submit their own work.
- Cite the original scientific paper and reused software.
- Disclose substantive AI assistance.
- Avoid fabricated references and experimental results.
- Use authorized testing environments.
- Protect personal and confidential data.
- Ensure that experiments are reproducible where feasible.

Using existing open-source projects is permitted when properly cited and meaningfully adapted.

Simply downloading and presenting an existing GitHub project without understanding, experimentation or adaptation is not acceptable.